

Splitting fields & Galois extensions

Thm M/K : finite extension

TFAE:

(1) M is Galois over K

(2) For all subextensions $M/L/K$

$$|\text{Emb}_K(L, M)| = [L:K]$$

(3) $\forall \alpha \in M$, with minimal poly,
 $m_\alpha(x) \in K[x]$ of deg d

$m_\alpha(x)$ has exactly d distinct
zeros in $M \Rightarrow m_\alpha(x)$ is separable

(4) $f(x) \in K[x]$ irred. of deg d

If $f(x)$ admits one zero in M
 then $f(x)$ has d distinct zeros in M

(5) There exists a separable
 polynomial $f(x) \in K[x]$
 of deg d
 s.t. (i) $f(x)$ has d distinct zeros in M
 $\alpha_1, \dots, \alpha_d \in M$
 (ii) $M = K(\alpha_1, \dots, \alpha_d)$

Pf.:

Proposition (1)

irred. $f(x) \in L[x]$

$$\begin{array}{c}
 M = L(\alpha) = L[x] / (f(x)) \\
 \left. \begin{array}{c} \downarrow \\ L \xrightarrow{\lambda} M, \lambda \in \text{Emb}_K(L, M) \\ \downarrow \\ K \end{array} \right\} \text{field extension}
 \end{array}$$

$$\text{Emb}_{L,2}(M, M) \stackrel{\text{defn}}{=} \{ \tilde{\lambda} \in \text{Emb}_K(M, M) : \}$$

$$\left(\begin{array}{l} \tilde{\lambda}|_L = 2 \\ (\exists) \tilde{\lambda}(\alpha) = 2(\alpha), \forall \alpha \in L \end{array} \right)$$

Then there is a bijection

$$\text{Emb}_{L,2}(M, M) \xrightarrow{\cong} \left\{ \alpha \in M : \begin{array}{l} \alpha \text{ is a} \\ \text{zero for} \\ \lambda(f(\alpha)) \in M[K] \end{array} \right\}$$

Pf: $M = L[x] / (f(x))$

$$\begin{array}{ccc} L[x] & \searrow \text{ev}_{2,\alpha} : \lambda(f)(\alpha) = 0 & \\ \downarrow & & \searrow \\ L[x] / (f(x)) & \xrightarrow{\quad} & M \end{array}$$

Example:

$$\begin{array}{ccc} \mathbb{Q}(i)(\sqrt{1+i}) & = & M = L[x] / (x^2 - (1+i)) \\ \text{deg } \sqrt{1+i} & | & | \\ \mathbb{Q}(i) & = & L \\ \text{deg } \sqrt{1+i} & | & | \\ \mathbb{Q} & = & K \\ \sqrt{1+i} & \xrightarrow{\quad} & \text{sq. rt of } (1+i) \end{array}$$

$$\begin{array}{ccc}
 \mathbb{M} & \xrightarrow{\lambda} & \mathbb{M} \\
 | & & || \\
 \mathbb{L} & \xrightarrow{\lambda} & \mathbb{M} \\
 | & \begin{array}{ccc} i & \mapsto & -i \\ 1+i & \mapsto & 1-i \end{array} & \\
 \mathbb{K} & &
 \end{array}$$

= root of
 $2(x^4 - (1+i))$
 $= x^4 - (1-i)$

In fact, $x^4 - (1-i)$ has no solutions in $\mathbb{M}$.

$$1-i = (-i)(1+i)$$

So having $\sqrt{1-i}$ in $\mathbb{M} \Leftrightarrow \sqrt{-i}$ in $\mathbb{M}$
 $\Leftrightarrow \sqrt{i}$ in $\mathbb{M}$.

$\Rightarrow \mathbb{M}$ is not Galois over $\mathbb{Q}$.

"Galois" is not a transitive property.

Corollary

 M $|$ L $|$ K

$$|\text{Emb}_K(M, M)|$$

$$\leq |\text{Emb}_K(L, M)| [M:L]$$

pf.: $M = \underbrace{L[f(x)]}_{(f(x))} \leadsto$ Proposition says

$$|\text{Emb}_K(M, M)| \leq |\text{Emb}_K(L, M)| \cdot \underbrace{\deg f(x)}_{[M:L]}$$

Reduce to the monogenic case via
induction

Pf: (1) $\Rightarrow$ (2)

Went:

M

$|$

L

$|$

K

$$|Emb_K(L, M)| \stackrel{\checkmark}{=} [L:K]$$

Heur's condition

$$|Aut(M/K)| \stackrel{d}{=} [M:K]$$

$\parallel$

$$|Emb_K(M, M)|$$

$$\stackrel{\text{Corollary}}{\leq} |Emb_K(L, M)| \cdot [M:L]$$

$$\stackrel{\text{red circle}}{\leq} [L:K] [M:L]$$

$$\leq [M:K]$$

$\Rightarrow$ All inequalities are equalities.

(2) $\implies$ (3)

$\alpha \in M$, M

$m_\alpha(x) \in K[x]$

$\uparrow$

degree d

$|$

$K(\alpha) = K[\alpha] = \underline{K[x]}$
($m_\alpha(x)$)

$|$

K

(2) $\implies | \text{Emb}_K(K(\alpha), M) | = \left| \left\{ \beta \in M : m_\alpha(\beta) = 0 \right\} \right|$

$\parallel$

$[K(\alpha) : K]$

$\parallel$

$d = \deg m_\alpha(x)$

$\implies m_\alpha(x)$ has d distinct zeros
in M

(3) $\Rightarrow$ (5)

$$M = K(\beta_1, \beta_2, \dots, \beta_m)$$

But then if we take

$$h(x) = m_{\beta_1}(x) m_{\beta_2}(x) \dots m_{\beta_m}(x)$$

then take $f(x)$ to be the product of the distinct irreducible factors of $h(x)$, then $f(x)$ will be a separate polynomial s.t.

M is generated by the zeros of $f(x)$ as an extension of K .

Defⁿ $f(x) \in K[x]$ non-constant.

A splitting field for $f(x)$ is a finite extension L/K s.t.

(1) $f(x) \in L[x]$ splits
into linear factors

$$f(x) = \prod_{i=1}^d (x - \alpha_i)$$

$$\alpha_i \in L$$

$$(2) L = K(\alpha_1, \dots, \alpha_d)$$

Reformulation of (5) of Theorem

M/K is Galois

$(\Rightarrow)$ M is a splitting

field for a separable

polynomial

